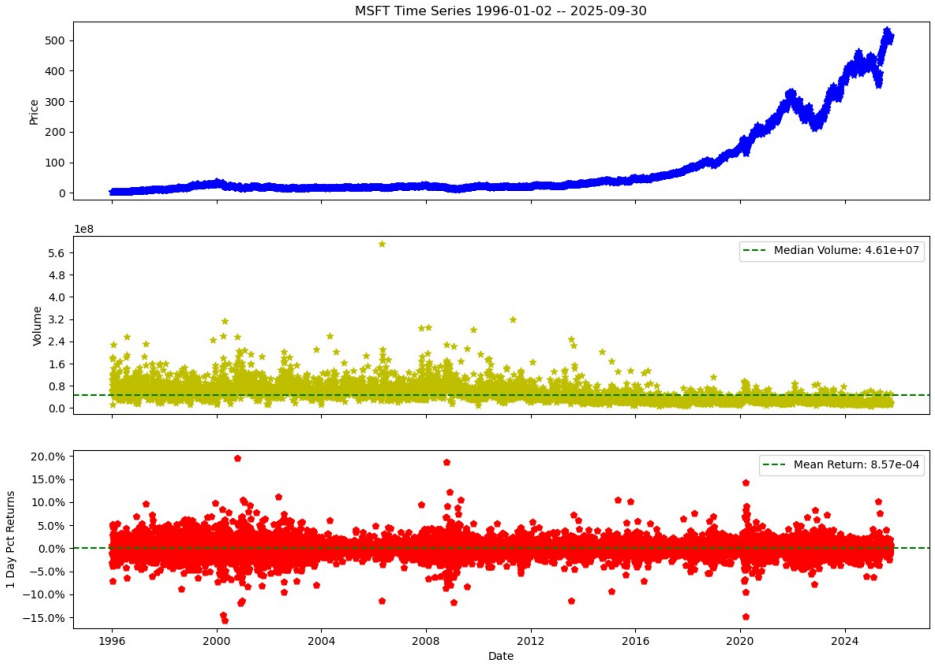
Module 6

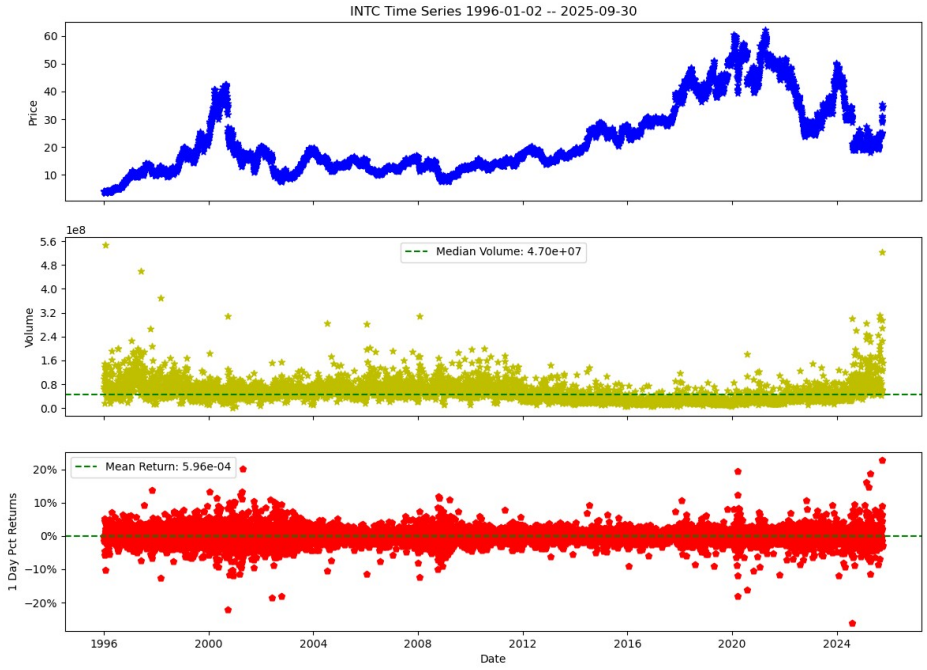
Option Pricing – Binomial Trees

October 1 & 6, 2025

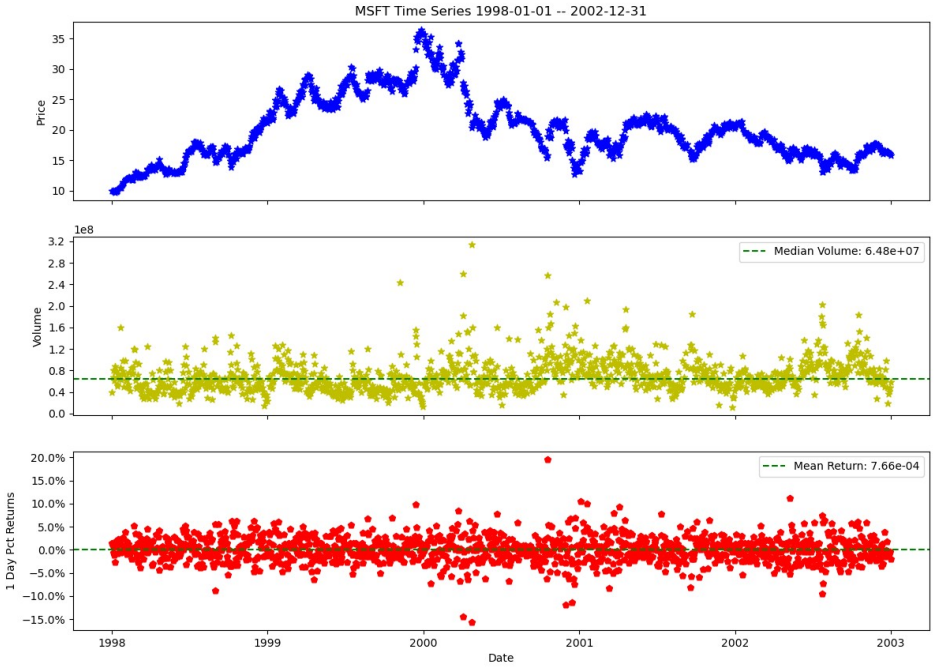
Stock Return and Return Volatility

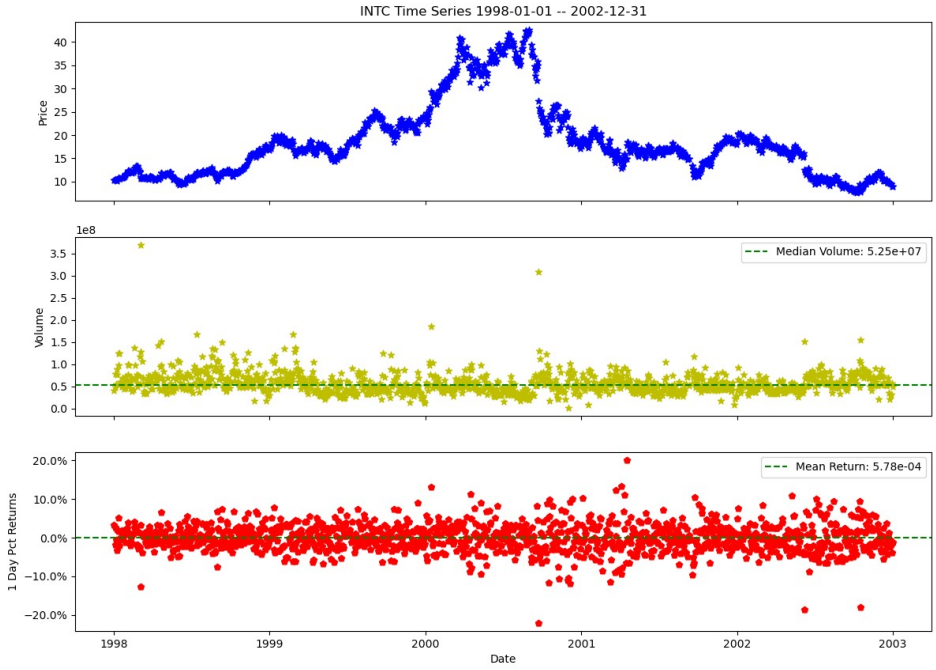
Samples of stock price time series (MSFT, INTC daily close, volume and % return) since 1996 (source: Yahoo Finance):



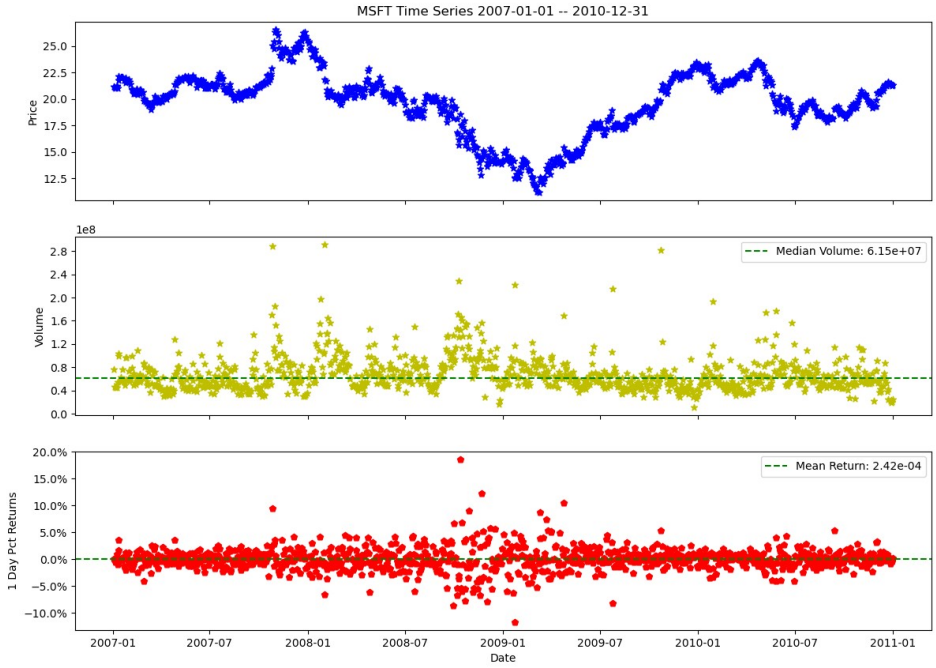

INTC Time Series 1996-01-02 -- 2025-09-30
Price
Volume
1e8
Median Volume: 4.70e+07
1 Day Pct Returns
Mean Return: 5.96e-04
Date

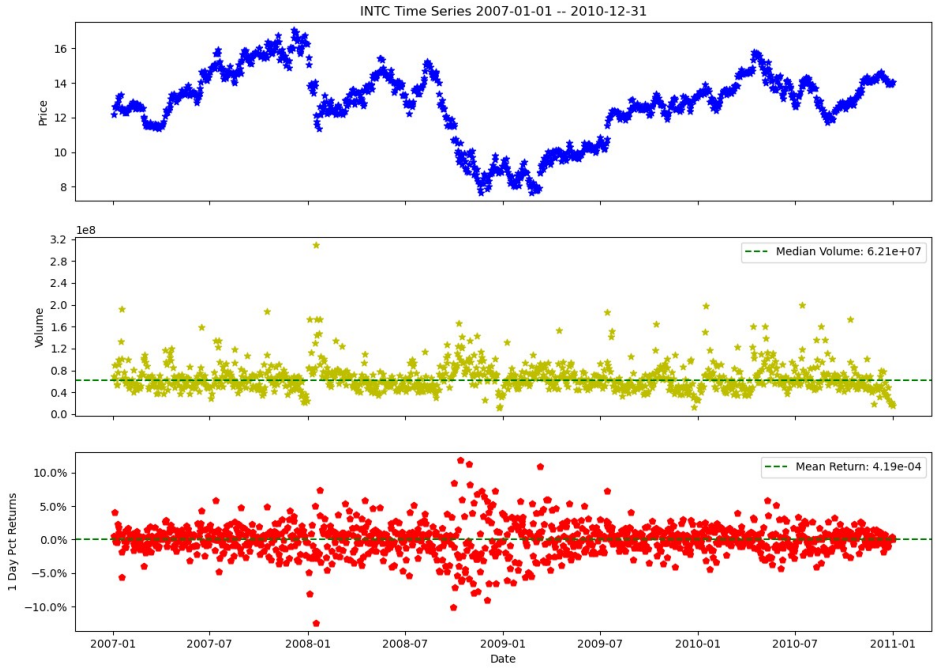
Dotcom Era:


MSFT Time Series 1998-01-01 -- 2002-12-31
Price
Volume
1e8
Median Volume: 6.48e+07
1 Day Pct Returns
Mean Return: 7.66e-04
Date

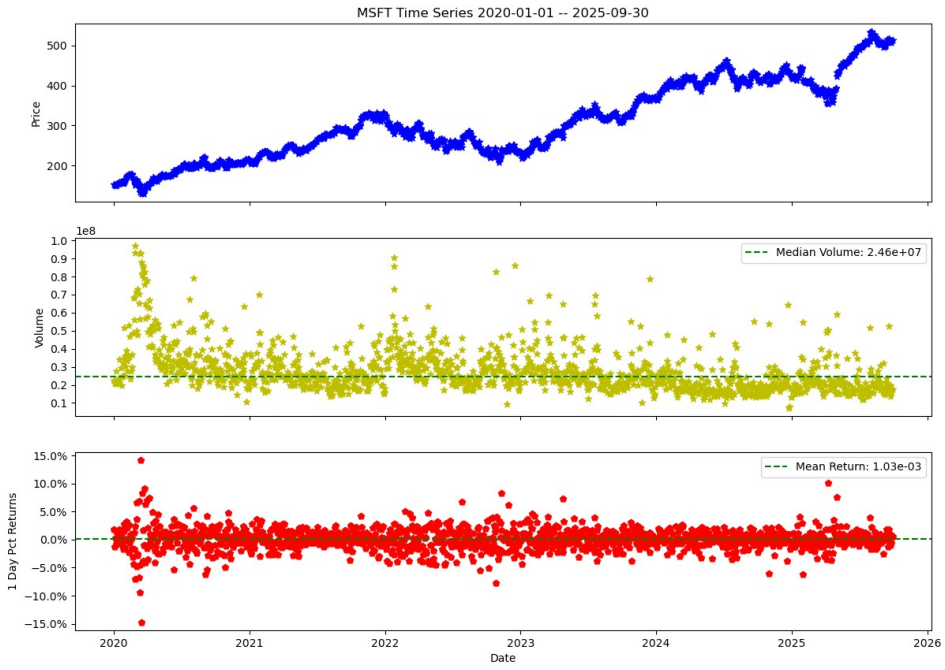


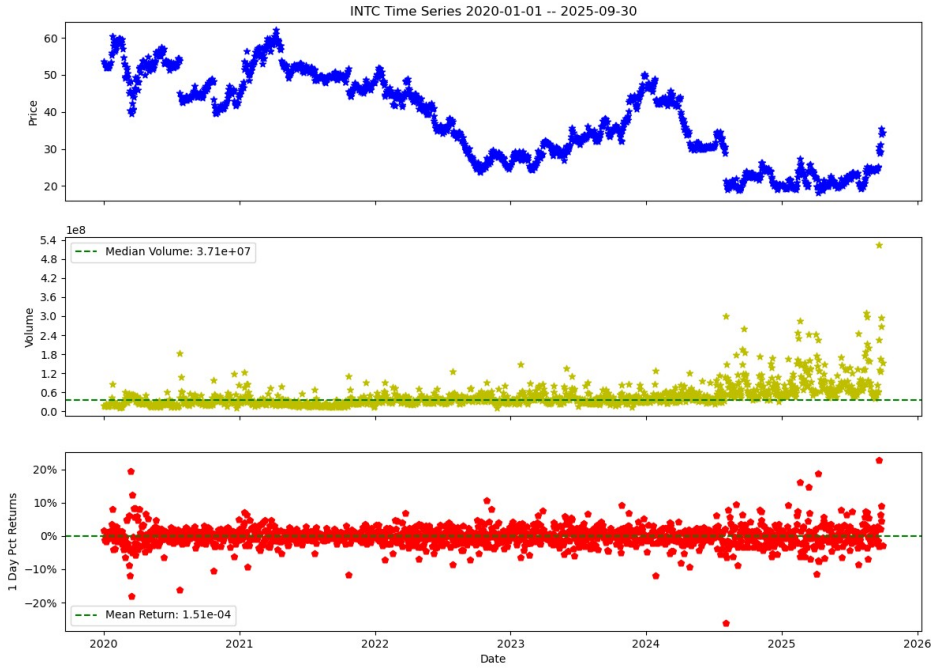
Great Recession:





Since Covid 19:





Key concepts:

Consider a time sequence $t_0, t_1, ..., t_n$ with share prices $S_0, S_1, ..., S_n$

- price vs return vs % return R:

  $$R_i = \frac{S_i}{S_{i-1}} - 1$$

  can be sampled at various frequencies: intraday, daily, weekly, …

  % return: random & independent

- average return $\bar{R}$

  $$\bar{R} = \frac{1}{n}\sum_{i=1}^{n} R_i$$

- uncertainty in return – variance:

  $$\text{var} = \frac{1}{n-1}\sum_{i=1}^{n}\left(R_i - \bar{R}\right)^2$$

- annualized standard deviation – "volatility":

  if S_i is sampled daily (R_i = daily return):

  $$\sigma = \sqrt{\frac{250}{n-1}\sum_{i=1}^{n}\left(R_i - \bar{R}\right)^2}$$

  250 is the annualization factor (US markets). For weekly return, it is 52, etc

  We need annualize volatility because t is measured in years

  We shall assume in the time period we are sampling { R_i } are independent and identically distributed normal random variables

  → constant volatility σ

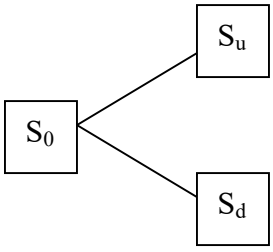
  → variance of return for a given period $\delta t = \sigma^2 \delta t$

We would like to construct a simple model to capture the random nature of stock price returns.

One Step Tree

Consider a call f mature at T. At maturity $f = \max(S_T - K, 0)$

Assume at t = 0: $S = S_0$; at t = T: S_0 → S_u , S_d (randomly), f → f_u, f_d



Let $S_0 = 100$, K = 105, T = 0.5 (6 months), r = 5%. Assume $S_u = 110$, $S_d = 90$,

→ at T: $f_u = 5$, $f_d = 0$

Valuation:

- Hedged portfolio: $\Delta S - f$ (for the option writer) Δ = # of shares in S

Select Δ such that $\Delta S - f$ has the same value at T, regardless the price of S_T

→ eliminate risk in $\Delta S - f$ → f

$$\Delta S_u - f_u = \Delta S_d - f_d \quad \rightarrow \quad \Delta = \frac{f_u - f_d}{S_u - S_d}$$

Δ = 5/20 = 0.25 → $\Delta S - f$ = 22.5 at T for all market scenarios;

at t = 0: $\Delta S - f$ = risk free discount of its time T value = 22.5 $e^{-0.025}$ = 21.9445

→ f = 0.25 × 100 – 21.9445 = 3.0555

- Replication: $A + \Delta S$, A = amount of cash

  Select A, Δ such that $A + \Delta S$ replicate f :

$$\begin{matrix} A_u + \Delta S_u = f_u \\ A_d + \Delta S_d = f_d \end{matrix} \quad , \quad A_u = A_d = Ae^{rT} = A_T \rightarrow \Delta = \frac{f_u - f_d}{S_u - S_d}, \quad A_T = \frac{S_u f_d - S_d f_u}{S_u - S_d}$$

$$\rightarrow \quad A = \frac{S_u f_d - S_d f_u}{S_u - S_d} e^{-rT}$$

$$\rightarrow \quad f = A + \Delta S = \frac{S_u f_d - S_d f_u}{S_u - S_d} e^{-rT} + \frac{f_u - f_d}{S_u - S_d} S_0$$

Δ = 0.25, A_T = -22.5 → A = -22.5 $e^{-0.0025}$ = -21.9445

f = 0.25 × 100 – 21.9445 = 3.0555

Arbitrage:

- $f_1 > f$: short the call at f_1, manufacture the same call at cost = f, pocket $f_1 - f > 0$
- $f_2 < f$: at t = 0, long the call at f_2, short Δ shares → certain cash flow at T = 22.5

  However PV(22.5) = 21.9445 > $f_2 - \Delta S$

  → Borrow cash $A_2 = f_2 - \Delta S$ at t = 0 to set up the long option, short stock position, pay it off at T, pocket the difference

General case:

- $S_u = uS_0$ $u > 1$ (u = 1.1 in the above example)
- $S_d = dS_0$ $d < 1$ (d = 0.9 in the above example)

$$f = \frac{f_u - f_d}{S_u - S_d} S_0 + \frac{S_u f_d - S_d f_u}{S_u - S_d} e^{-rT} = \frac{f_u - f_d}{u - d} + \frac{u f_d - d f_u}{u - d} e^{-rT}$$

$$= \frac{1 - d e^{-rT}}{u - d} f_u + \frac{u e^{-rT} - 1}{u - d} f_d = e^{-rT} \left[\frac{e^{rT} - d}{u - d} f_u + \frac{u - e^{rT}}{u - d} f_d \right]$$

$$= e^{-rT} \left[\frac{e^{rT} - d}{u - d} f_u + \left(1 - \frac{e^{rT} - d}{u - d} \right) f_d \right]$$

$$\rightarrow \; f = e^{-rT} \left[p f_u + (1 - p) f_d \right], \quad p = \frac{e^{rT} - d}{u - d}$$

Risk Neutral Valuation:

- If $u > e^{rT} > d$ → $0 < p < 1$ → p has a probability interpretation:

  $$E[S_T] = pS_u + (1 - p)S_d = S_0 e^{rT}$$

  p does not depend on the likelihood of S ends up or down

  → p is "risk neutral" probability

  $$f = e^{-rT} \left[p f_u + (1 - p) f_d \right] = e^{-rT} E[f_T]$$

  Pricing the above example under the risk neutral probability:

S0	100.0
K	105.0
T	0.5
r	5.0%

u	1.1
d	0.9
DF	0.9753
p	0.6266

European Call:

	110.00
	5.0000
Stk Px	100.00
call Px	3.0555
	90.00
	0.0000

Remarks:

- if $u < e^{rT}$: stock S upside is less than risk free rate + possibility of loss money at the downside → arbitrage opportunity → bad model

- if $e^{rT} < d$: stock S is guaranteed out perform risk free rate under all market scenario → arbitrage opportunity → bad model

- Real vs risk neutral world
- Does risk neutral probability always exist? → yes in "arbitrage free" models
- f does not depend on one's real world expectation of S, but it depends on u, d, the nature of the random process of S

Summary:

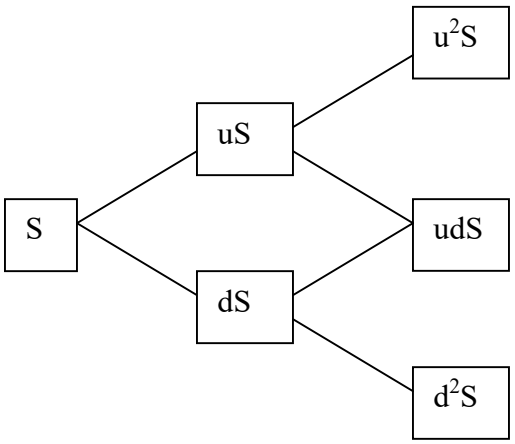
Three equivalent valuation paradigms:

1) Constructing a hedged portfolio that eliminate uncertainty in the final payoff
2) Replicating options with cash and underlying asset
3) Risk neutral pricing

Will they work beyond single step tree?

Two Step Trees

Trees are recombining by construction: udS = duS (dropped subscript 0)



One may repeat the above procedure at each node (i.e., uS, dS, then S), using any of the above 3 approaches. The risk neutral pricing scheme is the most convenient one.

Δ at each node:

$$\Delta_{uS} = \frac{f_{uu} - f_{ud}}{S_{uu} - S_{ud}}, \Delta_{dS} = \frac{f_{ud} - f_{dd}}{S_{ud} - S_{dd}}, \Delta_S = \frac{f_u - f_d}{S_u - S_d}$$

different at each node → dynamical hedging

Cash at each node (assume time step is δt):

$$A_{uS} = \frac{S_{uu} f_{ud} - S_{ud} f_{uu}}{S_{uu} - S_{ud}} e^{-r\delta t}, A_{dS} = \frac{S_{ud} f_{dd} - S_{dd} f_{ud}}{S_{ud} - S_{dd}} e^{-r\delta t}, A_S = \frac{S_u f_d - S_d f_u}{S_u - S_d} e^{-r\delta t}$$

also different at each node → dynamical replication

Look at node S_u = uS:

As $S \to S_u$, the hedging $\Delta_S \to \Delta_{uS}$ and cash $A_S \to A_S e^{r\delta t}$. The total new cash position resulted from adjusting stock position and interest accrual of the cash

$$A_{uS} = \left(\Delta_S S_u - \Delta_{uS} S_u\right) + A_S e^{r\delta t}$$

$$= \frac{f_u - f_d}{S_u - S_d} S_u - \frac{f_{uu} - f_{ud}}{S_{uu} - S_{ud}} S_u + \frac{S_u f_d - S_d f_u}{S_u - S_d} e^{-r\delta t} e^{r\delta t}$$

$$= \frac{f_u - f_d}{u - d} u - \frac{f_{uu} - f_{ud}}{u - d} + \frac{u f_d - d f_u}{u - d}$$

$$= f_u - \frac{f_{uu} - f_{ud}}{u - d}$$

$$= \frac{f_{uu} - f_{ud}}{S_{uu} - S_{ud}} S_u + \frac{S_{uu} f_{ud} - S_{ud} f_{uu}}{S_{uu} - S_{ud}} e^{-rT} - \frac{f_{uu} - f_{ud}}{u - d}$$

$$= \frac{S_{uu} f_{ud} - S_{ud} f_{uu}}{S_{uu} - S_{ud}} e^{-rT}$$

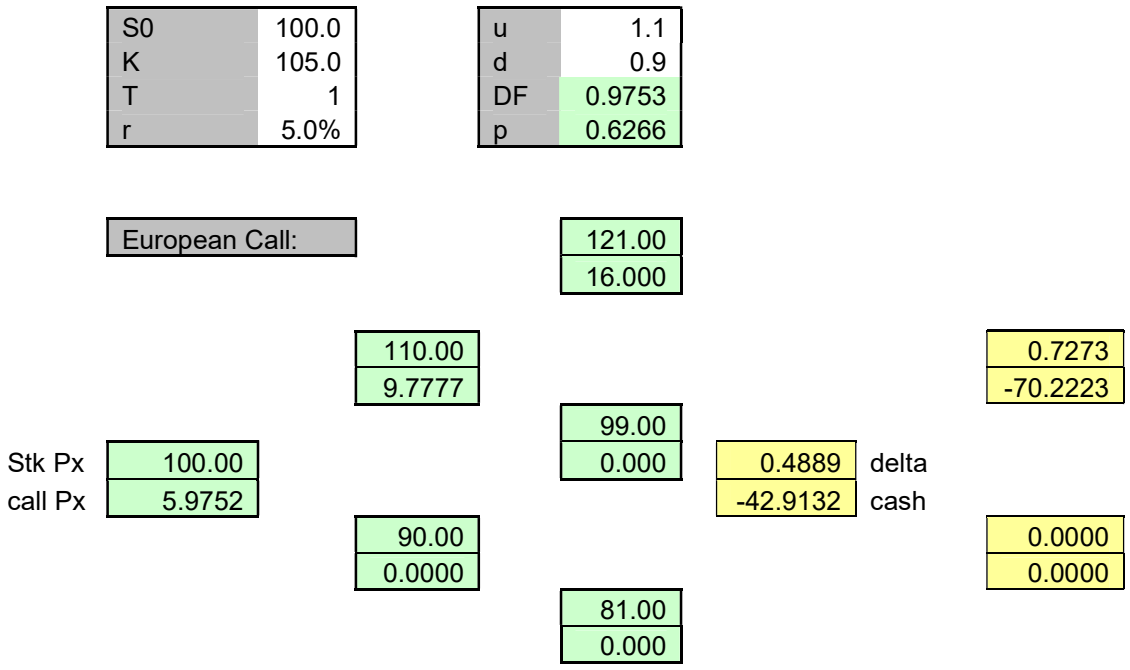
A_{uS} has the same expression as previously obtained by walking backward

Similarly for the S_d = dS node

→ No net new funding is needed, no fund is taken away

→ rebalancing between stock and cash is "self financing"

Example: let $S_0 = 100$, K = 105, T = 1.0 (1 year), r = 5%. Assume u = 1.1, d = 0.9



- How should one hedge?

  Assume short call:

  $t = t_0 = 0$:

  - long 0.4889 share at S = 100

  $t = t_1 = 0.5$:

  - if S = 110, hedging position = 0.4889 × 110, long additional shares so that after rebalancing one long 0.7273 shares at 110

  - if S = 90, sell all shares

  $t = t_2 = T = 1$:

  if stock was 110 when $t = t_1$:

  - if S = 121, hedging position = 0.7273 × 121 = 88, option = 16, total = 72

  - if S = 99, hedging position = 0.7273 × 99 = 72, option = 0, total = 72

  if stock was 90 when $t = t_1$:

  - if S = 99, hedging position = 0, option = 0, total = 0

  - if S = 81, hedging position = 0, option = 0, total = 0

→ uncertainty at option maturity removed

But what is the cost of hedging?

Along the S = 100 → 110 paths:

$0.4889 \times 100 \times e^{0.05} + (0.7273 - 0.4889) \times 110 \times e^{0.025} - 72 = 6.8216$

$6.8216 \times e^{-0.05} = 5.9752$

Along the S = 100 → 90 paths:

$0.4889 \times 100 \times e^{0.025} - 0.4889 \times 90 = 6.1265$

$6.1265 \times e^{-0.025} = 5.9752$

→ option hedging cost = option fair price

- How should one replicate?

  $t = t_0 = 0$:

  - long 0.4889 share at S = 100, borrow 42.9132

  $t = t_1 = 0.5$:

  - if S = 110, stock position = 0.4889×110, long additional shares so that after rebalancing one long 0.7273 shares at 110. Funding required = $(0.7273 - 0.4889) \times 110 = 26.2227$. Total borrow = $42.9132 \times e^{0.025} + 26.2227 = 70.2223$

  - if S = 90, sell all shares to receive = $0.4889 \times 90 = 43.9996$. Use it to pay off debt = $42.9132e^{0.025} = 43.9996$. Position value = 0

  $t = t_2 = T = 1$:

  if stock was 110 when $t = t_1$:

  - if S = 121, stock position = $0.7273 \times 121 = 88$, debt = $70.2223e^{0.025} = 72$ total position value = 16

  - if S = 99, stock position = $0.7273 \times 99 = 72$, debt = $70.2223e^{0.025} = 72$, total position value = 0

  if stock was 90 when $t = t_1$:

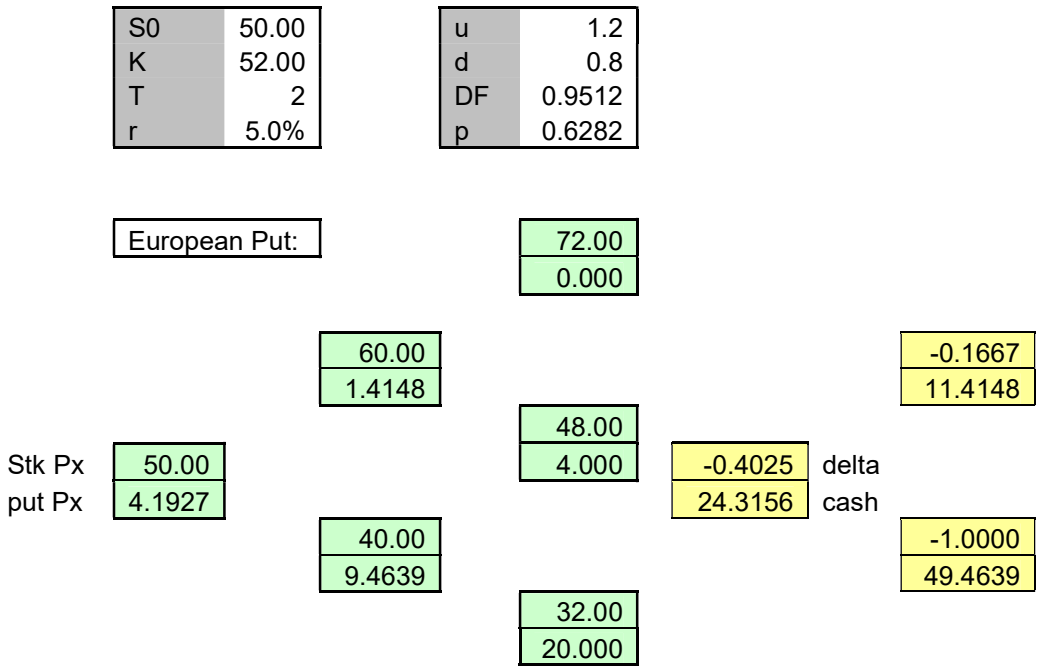
  - if S = 99, stock position = 0, cash = 0, total = 0

  - if S = 81, stock position = 0, cash = 0, total = 0

  → option payoff fully replicated

  Cost of setting up the initial position = $0.4889 \times 100 - 42.9132 = 5.9752$ = option price

Put examples:



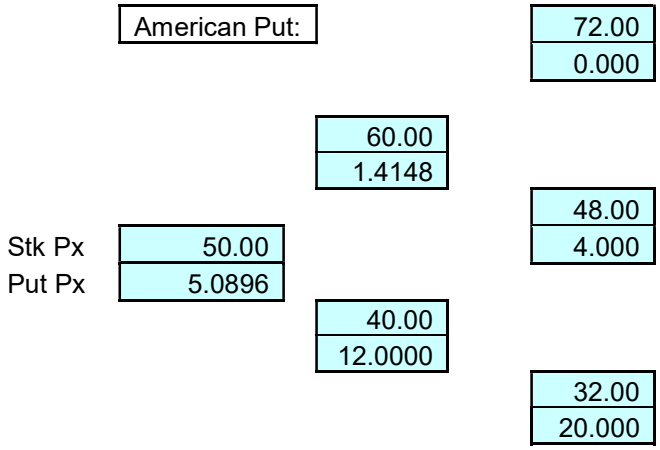
American Option:

Check early exercise at each node:

- Call: max($S_t - K, C_t$)
- Put: max($K - S_t, P_t$)

American Put:

In the above example, European → American



Determining u and d:

- Matching stock return volatility:

  Consider a small time increment δt where $S \to S_u, S_d$:

  $$S_u = uS, S_d = dS \rightarrow R_u = \frac{S_u}{S} - 1 = u - 1, R_d = \frac{S_d}{S} - 1 = d - 1$$

  Return variance:

  $$\text{var} = \sigma^2 \delta t$$

  $$= p(u-1)^2 + (1-p)(d-1)^2 - \left[p(u-1) + (1-p)(d-1)\right]^2$$

  $$= p(1-p)\left[(u-1) - (d-1)\right]^2$$

  $$= p(1-p)(u-d)^2$$

  $$p = \frac{e^{r\delta t} - d}{u - d}, \quad 1 - p = \frac{u - e^{r\delta t}}{u - d}$$

  $$\rightarrow (e^{r\delta t} - d)(u - e^{r\delta t}) = \sigma^2 \delta t$$

  Let d = 1/u:

  $$(e^{r\delta t} - d)(u - e^{r\delta t}) = (u + d)e^{r\delta t} - e^{2r\delta t} - 1 = \sigma^2 \delta t$$

  $$u + d = e^{-r\delta t} + e^{r\delta t} + \sigma^2 \delta t e^{-r\delta t} = 2 + \sigma^2 \delta t + O(\delta t^2)$$

  Let $u = e^a$,

  $$e^a + e^{-a} = 2 + \sigma^2 \delta t \quad \rightarrow \quad a^2 = \sigma^2 \delta t + O(\delta t^2)$$

  $$\rightarrow u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}}$$

  p is risk neutral probability, σ is real world volatility

- Volatility in risk neutral and real world matches (Girsanov's theorem) :

  Assume in real world stock has return rate μ and probability p^*,

  $$E^*[S] = p^* S_u + (1 - p^*) S_d = S_0 e^{\mu\delta t}$$

  $$p^* = \frac{e^{\mu\delta t} - d}{u - d}$$

$\rightarrow u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}}$

Volatility: a key variable that affects option value!

N Step Trees

Cook book:

- Set tree building parameters (Cox-Ross-Rubinstein (CRR) model):

  $\delta t = \frac{T}{N}, \; p = \frac{e^{r\delta t} - d}{u - d}, \; u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}}$

- Grow the N -step tree for S:

  $0 = t_0 \rightarrow t_1 = t_0 + \delta t \rightarrow t_2 = t_1 + \delta t \rightarrow ... \rightarrow t_N = T$

  $S \rightarrow (uS, dS) \rightarrow \left(u^2 S, udS, d^2 S\right) \rightarrow \left(u^3 S, u^2 dS, ud^2 S, d^3 S\right) \rightarrow ...$

  Assume the final branch is $S_0^{t_N} < S_1^{t_N} < .... < S_N^{t_N}$

- Calculate the payoff at the terminal branch $t_N = T$:

  - Call: $f_i^{t_N} = \max(S_i^{t_N} - K, 0)$
  - Put: $f_i^{t_N} = \max(K - S_i^{t_N}, 0)$

- Backward induction:

  $f_i^{t_j} = e^{-r\delta t}\left[p f_{i+1}^{t_{j+1}} + (1-p) f_i^{t_{j+1}} \right]$

  Check possible early exercise for American options

Exercise:

Let S = 100, r = 5.5%, σ = 35%. S pays no dividend. Price a European and an American put option with K = 95, T = 1.0 using a N = 100 binomial tree described above.

Answer: p = 8.7638, P = 9.1932

(We will have a HW on general construction of binomial tree later in the course)

Note:

Alternative Tree Constructions:

- CRR:

$$u = \mathrm{e}^{\sigma\sqrt{\delta t}}$$
$$d = 1/u$$
$$p = \frac{e^{(r-q)\delta t} - d}{u - d}$$

- CRR Equal Probability:

$$u = \mathrm{e}^{\left(r-q-\frac{\sigma^2}{2}\right)\delta t+\sigma\sqrt{\delta t}}$$
$$d = \mathrm{e}^{\left(r-q-\frac{\sigma^2}{2}\right)\delta t-\sigma\sqrt{\delta t}}$$
$$p = \frac{1}{2}$$

- Jarrow & Rudd:

$$u = \mathrm{e}^{\sigma\sqrt{\delta t}}$$
$$d = 1/u$$
$$p = \frac{1}{2} + \frac{r - q - \frac{\sigma^2}{2}}{2\sigma}\sqrt{\delta t}$$

q : continuous dividend yield